Main-text figure redesigns

Eight figures, four behavioral comparison options | 30 September 2026

1. Temporary design drafts; manuscript source and production figures are unchanged.

2. Figure 2: the recovered plot contains 25 participant values per pattern. Their equal-weight mean is 82.4708%; the old pooled-trial human line averages 86.7976%. Point indices are not participant IDs, and no paired cross-pattern analysis is inferred from their order.

3. Figure 3A-C preserve the exact current scatterplot points to isolate layout choice. Figure 3D is a separately labeled sensitivity calculation using the equal-participant profile from Figure 2. It changes correlations (e.g., Phi-4: .688 to .784; Llama-70B: .266 to .426). Choose and document the behavioral estimand before publication.

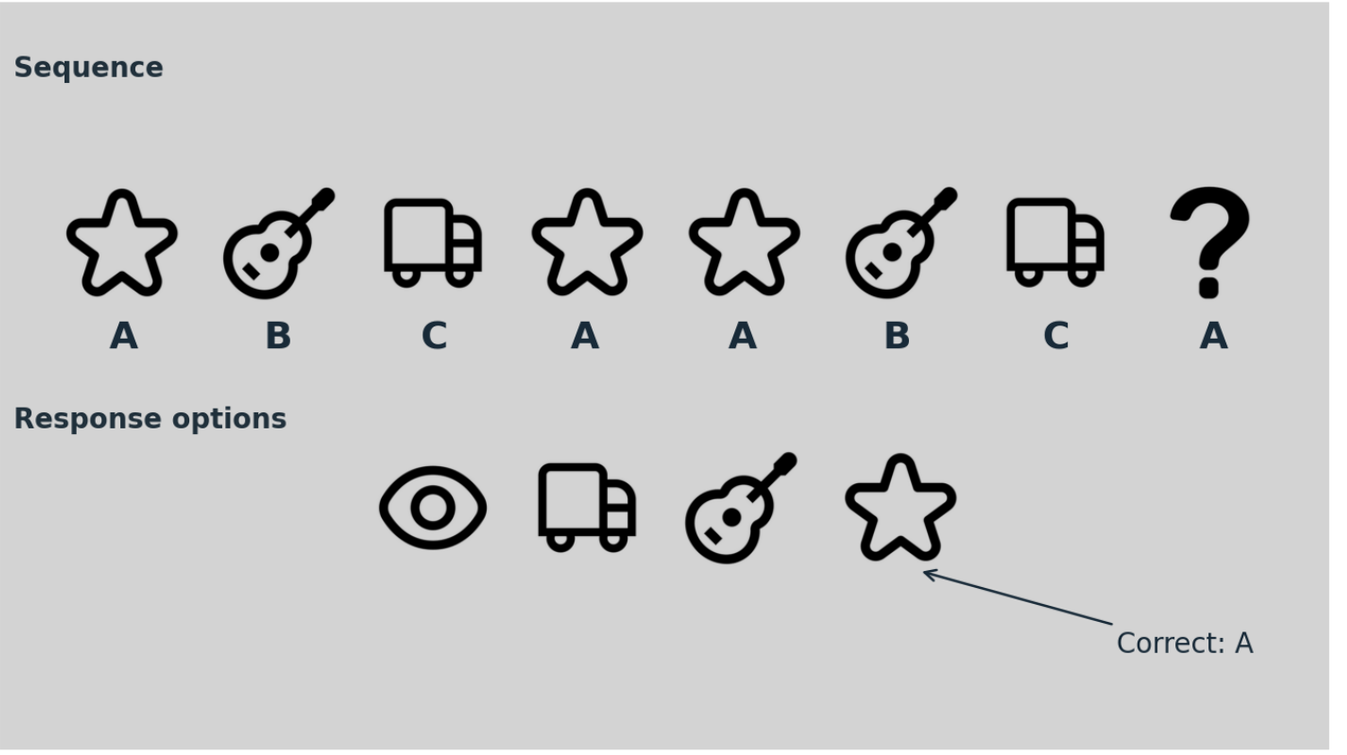
4. Figures 6-7 both use matched participants/sessions. The original Figure 7 used condition-specific sets. This accounts for the response mean discrepancy (.285 versus .192). The draft is harmonized to Figure 6 and the matched comparison described in the text.

5. Figure 8 retains the saved Fisher-z estimator and existing participant confidence intervals. It does not silently switch to the arithmetic mean of raw correlations.

6. All plot labels and marks are vector in PDF. Figure 1 retains the original stimulus raster; the 400-by-400 empirical RDM is rasterized without interpolation. No new raw-data cleaning or model inference was run.

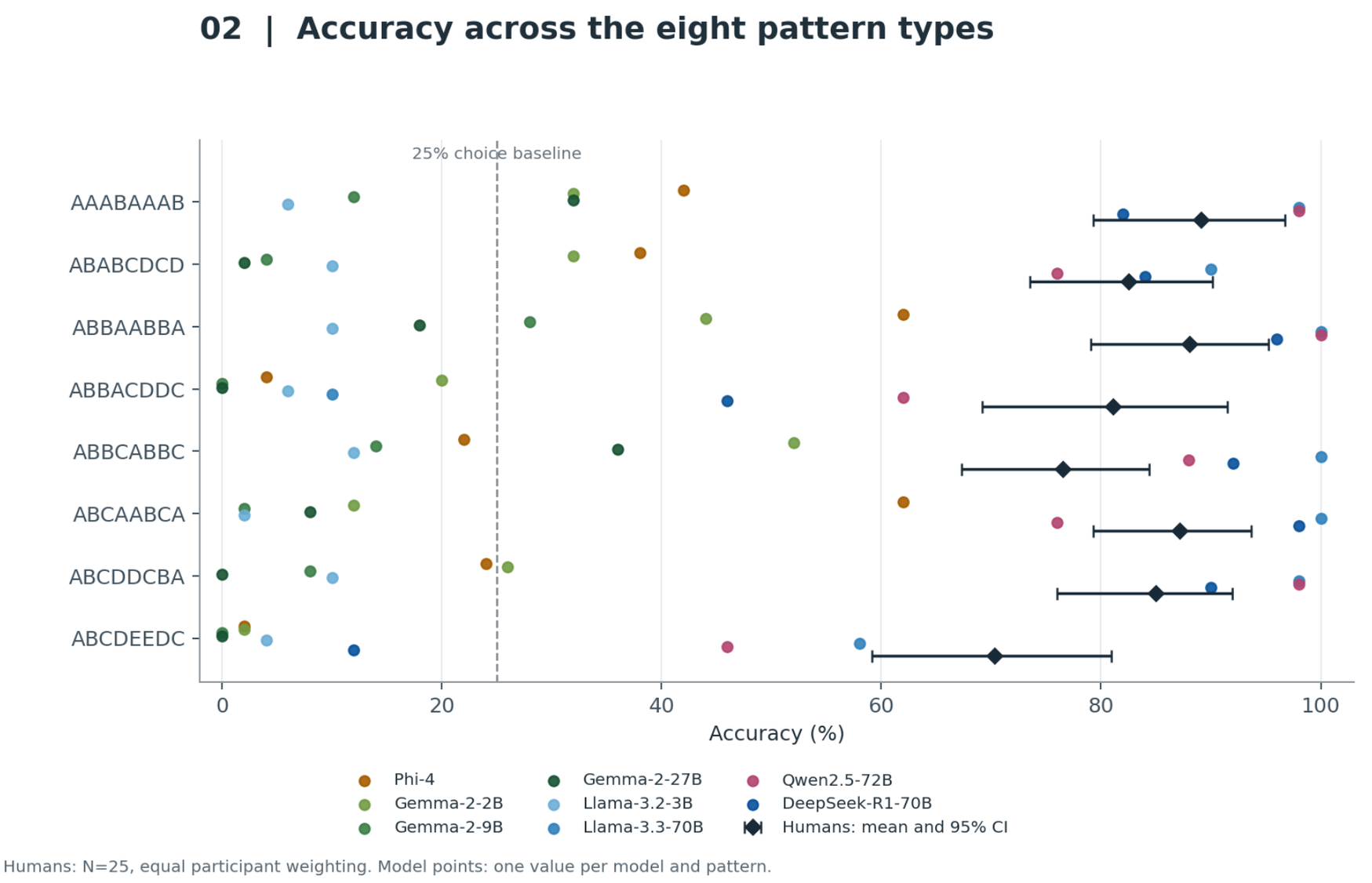
7. DeepSeek-R1-70B abbreviates DeepSeek-R1-Distill-Llama-70B throughout. Each output has a PNG preview and source data in data/.

01 | A worked example of the task



Letters, answer annotation and timeline are explanatory additions; the stimulus image is retained.

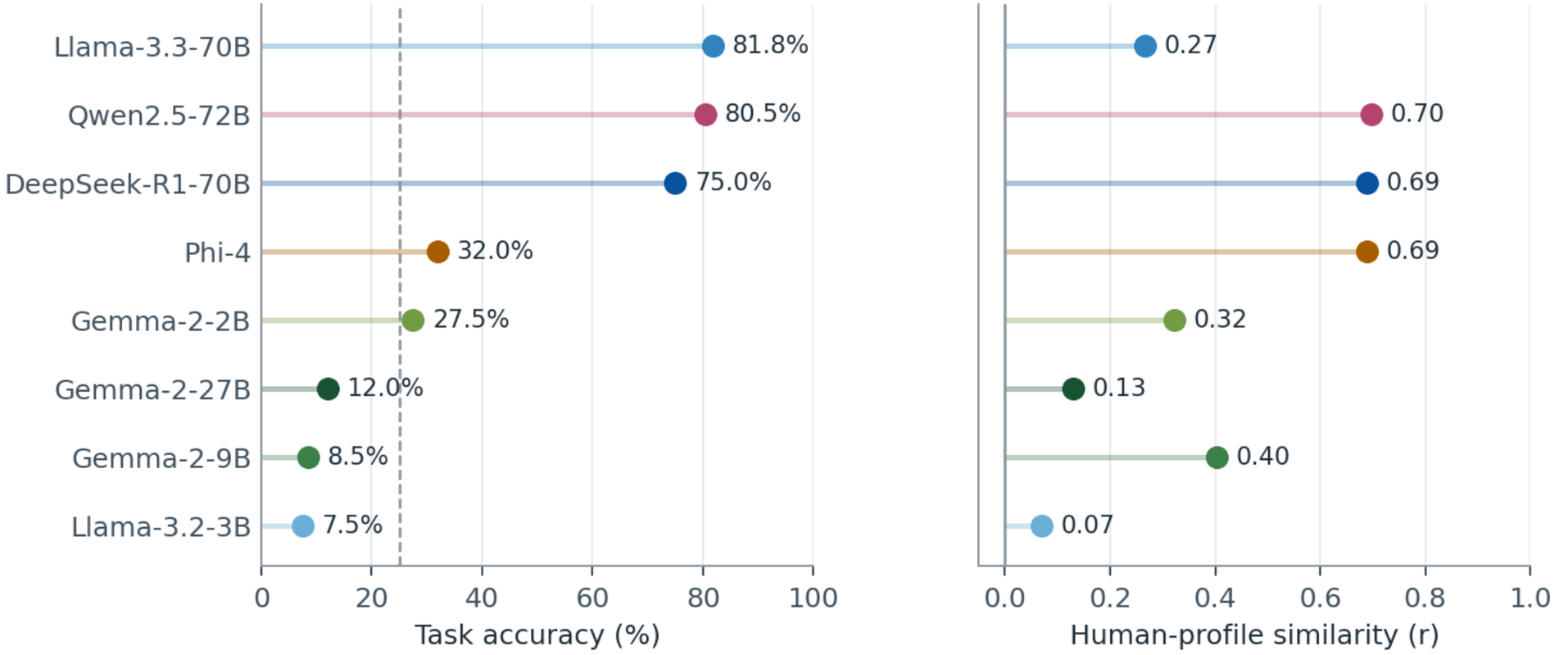
Original stimulus with pattern letters, correct-answer annotation and a schematic fixation-locked EEG window. The letters and annotations are not part of the experimental display. Image content is raster; annotations are vector.



Human means give each of 25 participants equal weight. Whiskers are marginal 95% percentile bootstrap CIs (10,000 resamples of participant values within each pattern; seed 20260930). Model points are descriptive accuracies, without sampling intervals. The 25% line is the uniform four-choice baseline; invalid model outputs can produce below-baseline accuracy. Overall human mean across patterns: 82.47%.

03 | Accuracy and human-like difficulty are distinct

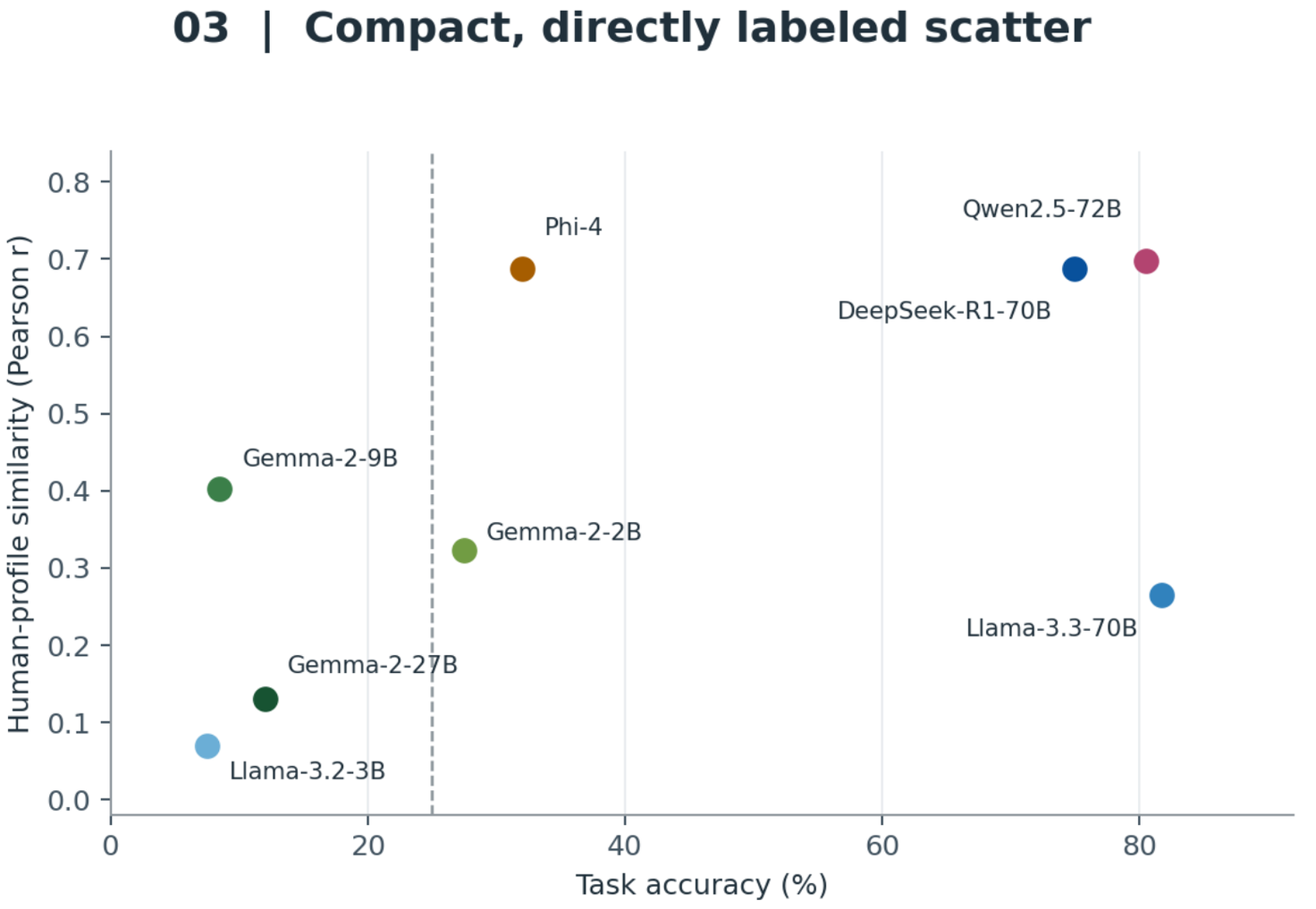
Layout comparison: values preserved from the current manuscript scatterplot.



Models ordered by accuracy. The two columns use separate scales; intervals are not estimated.

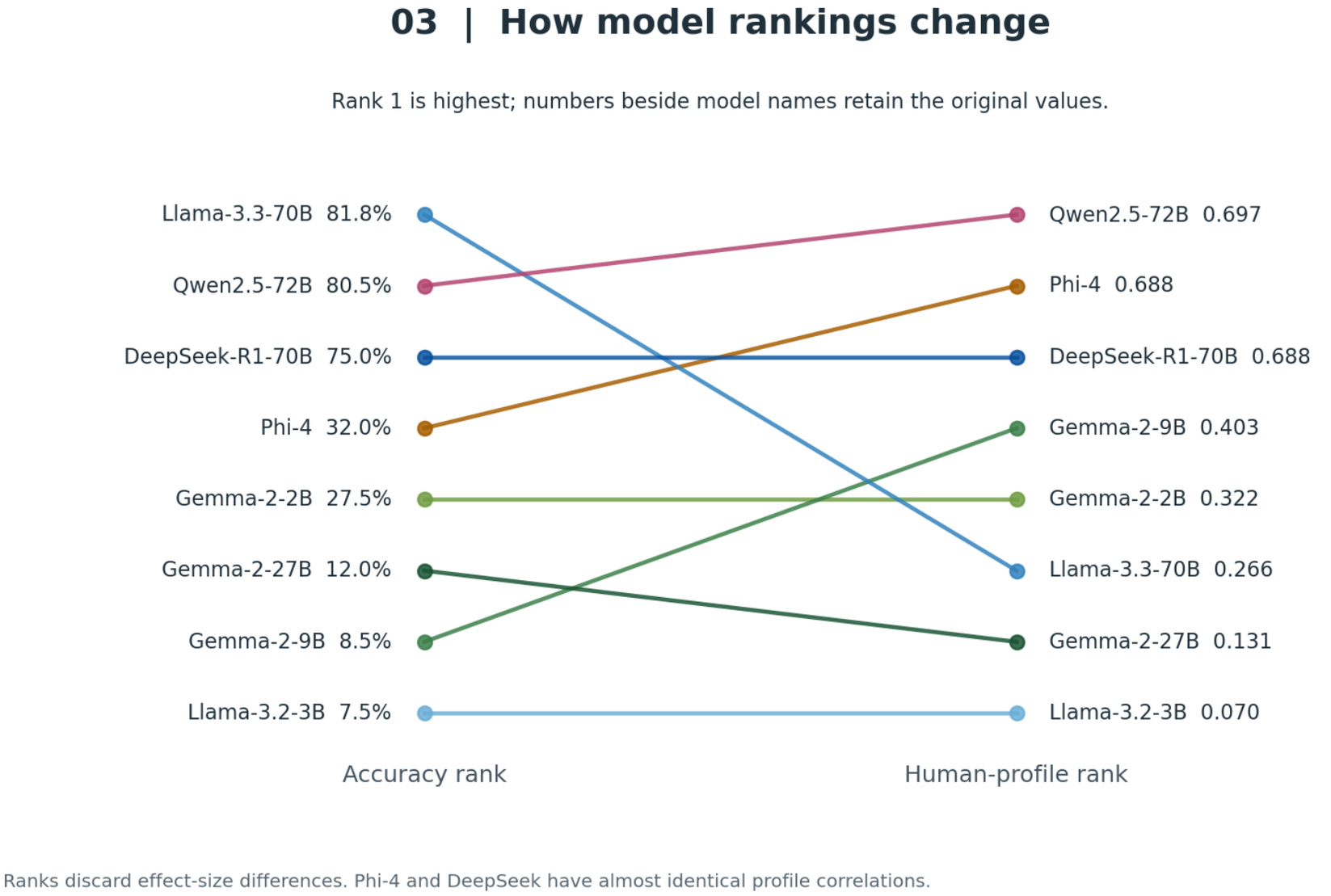
Aligned dot plots, ordered by accuracy. Layout comparison: values preserved from the current manuscript scatterplot. Exact values are annotated. Separate axes avoid implying that accuracy and correlation are the same quantity.

03b accuracy profile compact scatter



Compact scatter with direct model labels. Current manuscript values are preserved. The vertical axis uses the displayed range around the data, rather than the full possible correlation range. This remains the clearest option for judging the relationship between the two metrics.

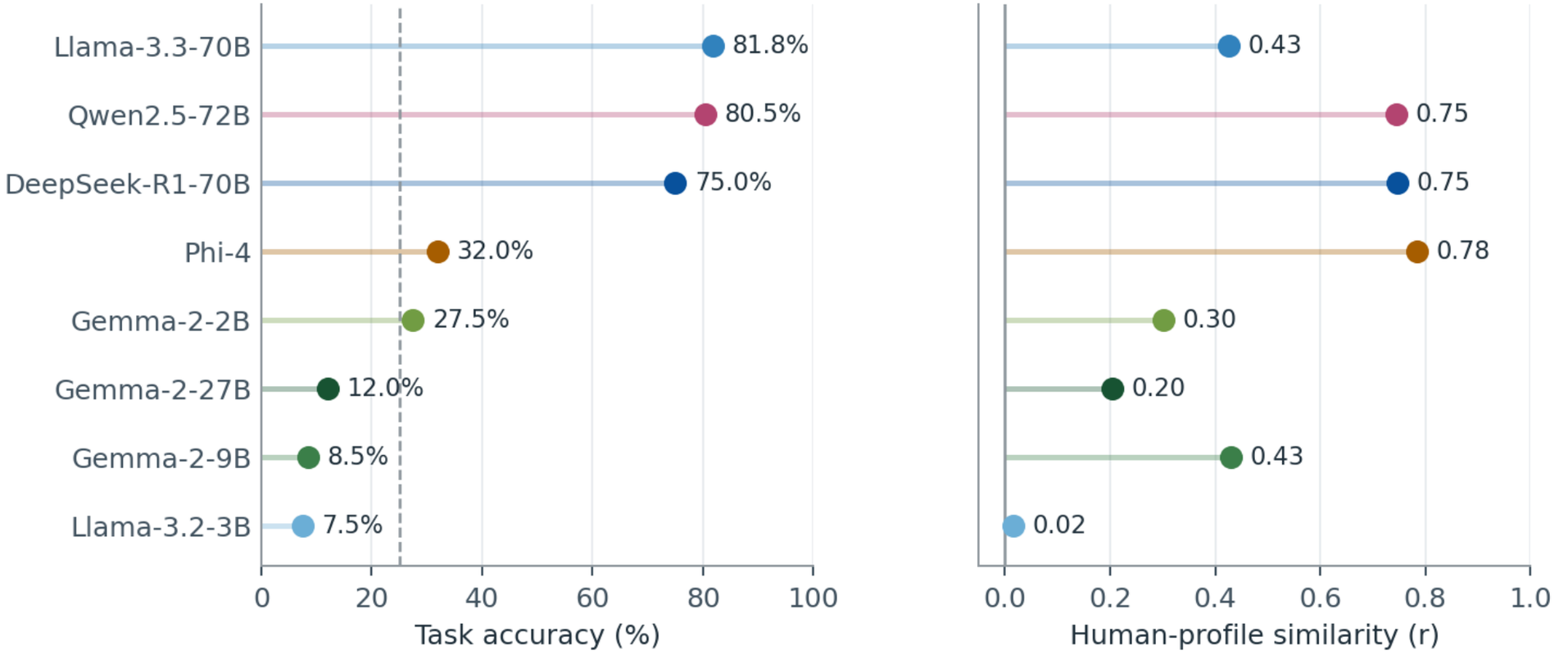
03c accuracy profile rank



Rank comparison using current manuscript values. It emphasizes changes in ordering, but overstates tiny rank differences if interpreted without the printed values: Phi-4 and DeepSeek correlations differ by about 0.00006. This is a descriptive alternative, not evidence of statistically distinguishable model ranks.

03 | Accuracy and human-like difficulty are distinct

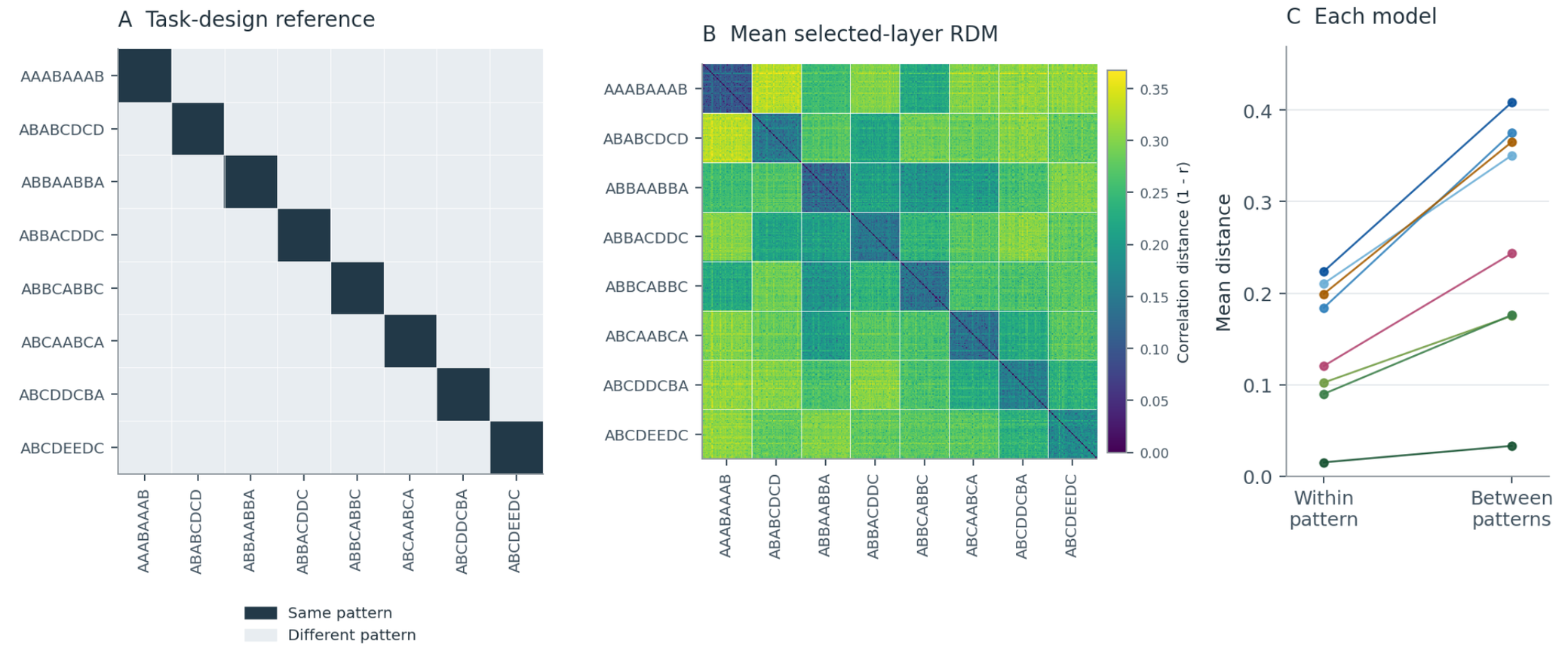
Analysis sensitivity: correlations recomputed against Figure 2's equally weighted human profile.



Models ordered by accuracy. The two columns use separate scales; intervals are not estimated.

Aligned dot plots, ordered by accuracy. Analysis sensitivity: correlations recomputed against Figure 2's equally weighted human profile. Exact values are annotated. Separate axes avoid implying that accuracy and correlation are the same quantity.

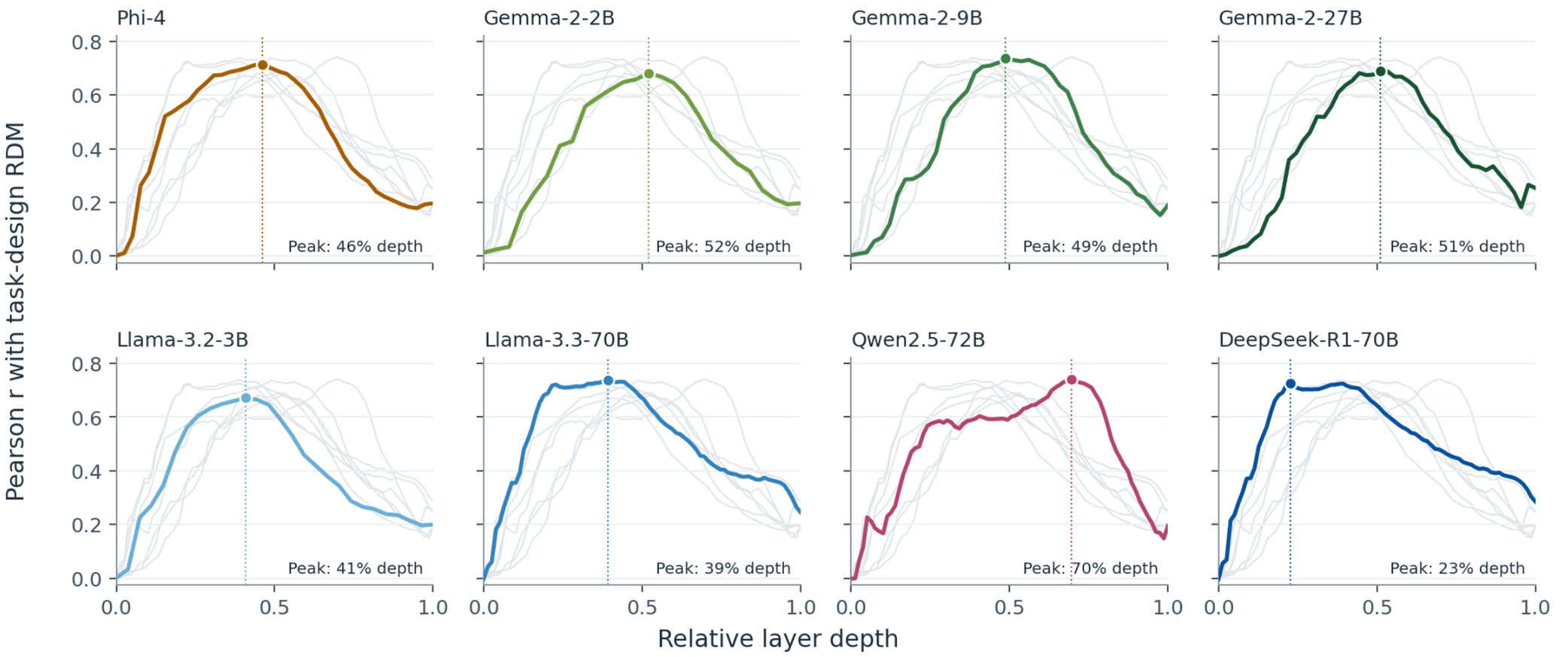
04 | Trial representations cluster by abstract pattern



400 trials; 50 per pattern. Panel C: one connected pair per model; self-distances excluded.

Reference uses a binary same/different-pattern key; measured LLM RDM uses correlation distance. Panel C averages unique within- and between-pattern pairs separately for each of eight models; these are descriptive summaries, not independent pair-level inferential tests. The full trial matrix is preserved, not replaced by an RDM of pattern-averaged features.

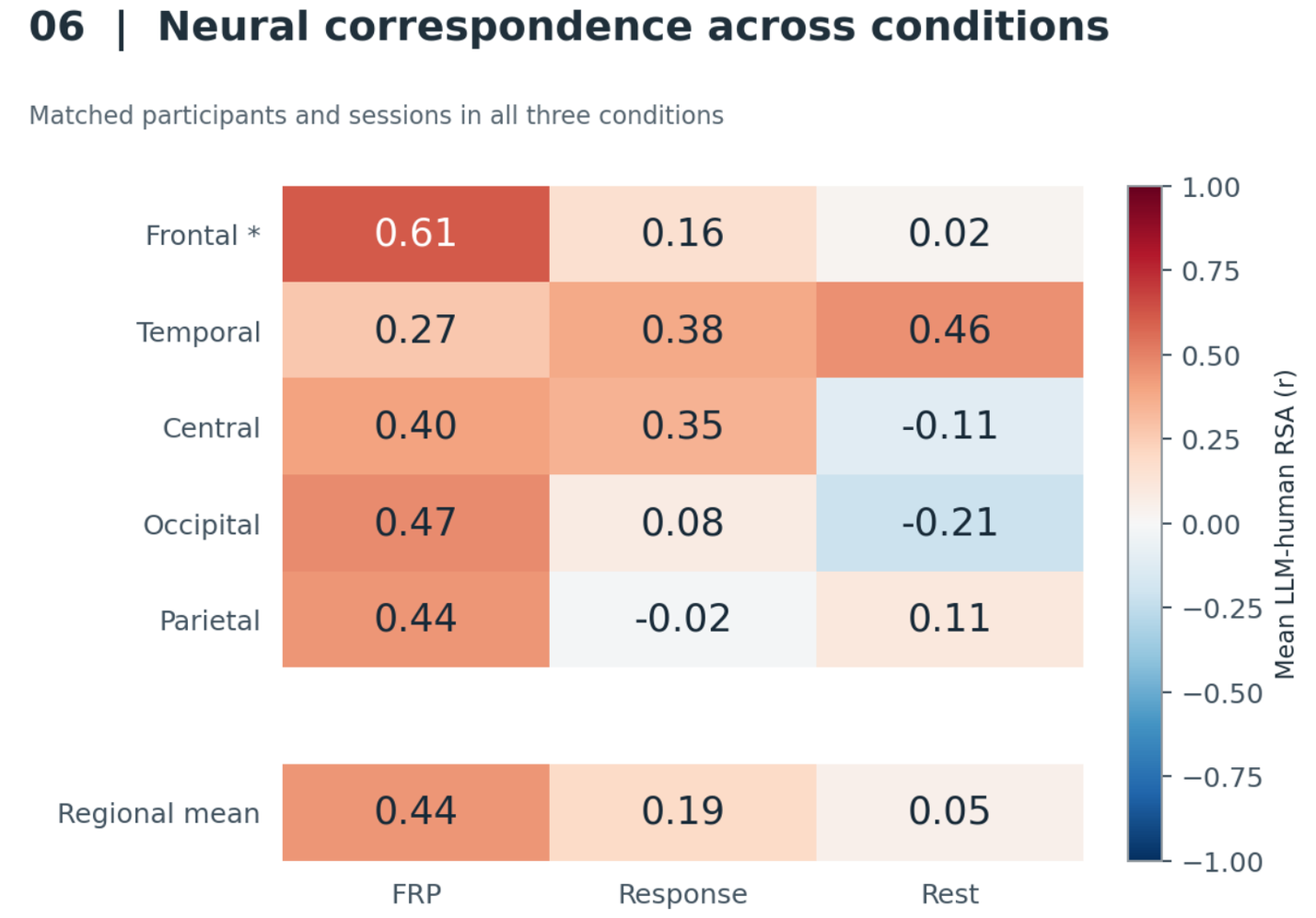
05 | Task structure across model depth



Colored line: named model. Gray lines: other models. Dot: task-selected layer, independent of EEG.

One panel per model with shared axes. Gray trajectories provide context; the colored dot marks the maximum task-design similarity. Normalized depth follows the existing figure convention: layer index divided by the maximum index, not an independently redefined depth measure.

06 matched rsa heatmap



Same 24 participants and 96 complete participant-session cells across conditions. Each regional cell is the arithmetic mean of eight model RSA values. The regional mean is separated from anatomical regions. Frontal is marked as primary. No cellwise significance is implied.

07 | Model correspondence and regional variation

Same matched dataset as Figure 6; model order follows task accuracy

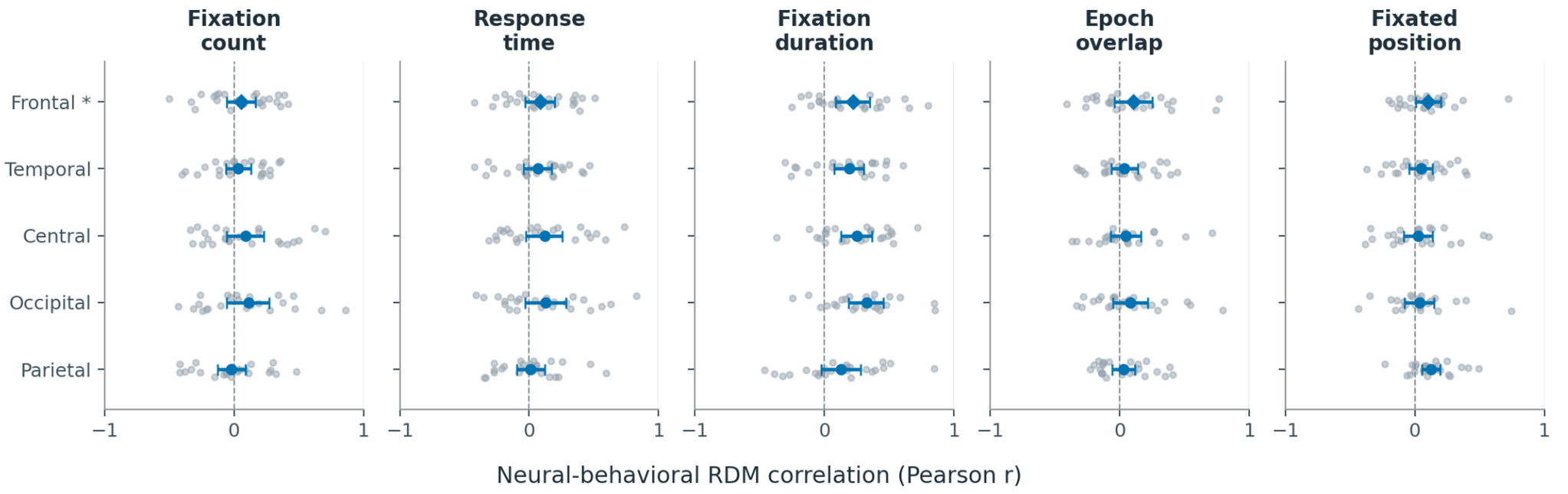


Small points are regional values, not participant observations or confidence intervals.

Five regional values per model and condition; frontal values are diamonds and regional means are open squares. Uses exactly the matched data underlying Figure 6. This deliberately replaces the condition-specific participant sets used by the previous Figure 7, so response means change from about .285 to .192. The five regions are not treated as independent participant replicates.

08 | FRP associations with gaze and timing

Participant values and 95% confidence intervals reveal variation hidden by the heatmap



N=24. Blue: Fisher-z mean and t-based 95% CI, back-transformed to r. Gray: participants. * Primary region.

Participant dots and the existing Fisher-z mean / t-based 95% confidence intervals, back-transformed to r. N=24 in every region-measure cell. Intervals are marginal, not simultaneous or multiplicity-adjusted. The source heatmap also uses Fisher-z summaries; these are preserved. Associations do not establish that gaze/timing explains away LLM-FRP correspondence.